

Grade 7 ICT – Central Processing Unit (CPU) Notes

Central Processing Unit (CPU)

The **Central Processing Unit (CPU)** is known as the **brain of the computer**. It is a digital circuit that processes data according to given instructions and converts data into useful information.

Example

A computer is like a **hotel kitchen**, and the **CPU is like the chef**.

- The chef prepares food.
 - The CPU processes data.
 - A faster chef prepares food more quickly.
 - A faster CPU processes data more quickly.
 - A **dual-core processor** is like having **two chefs** working at the same time, allowing two tasks to be processed simultaneously.
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Location of the CPU

- The CPU is **inside the system unit**.
- It is **fixed on the motherboard**.
- It **cannot be seen from outside** the computer.

System Unit → Motherboard → CPU

Functions of the CPU

The CPU performs the following tasks:

- Processes data according to instructions.
 - Executes computer programs.
 - Controls all computer operations.
 - Converts data into information.
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Main Components of the CPU

The CPU has **three main components**:

1. Arithmetic and Logic Unit (ALU)

The **ALU** performs **mathematical** and **logical** operations.

Mathematical Operations

- Addition (+)
- Subtraction (−)
- Multiplication (×)
- Division (÷)

Example:

- $8 + 5 = 13$
- $20 \div 4 = 5$

Logical Operations

Compares two values.

Examples:

- $5 > 2$
 - $10 = 10$
 - $7 < 9$
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2. Control Unit (CU)

The **Control Unit** controls and coordinates all activities inside the computer.

Functions

- Controls all hardware devices.
 - Controls input and output operations.
 - Sends data to the correct place at the correct time.
 - Checks whether signals are received correctly.
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3. Memory Registers

Memory Registers are **very small, high-speed temporary memory locations** inside the CPU.

Functions

- Temporarily store data and instructions currently being processed.
- Access data much faster than hard disks or other storage devices.
- Have very small storage capacity.

Summary of CPU Components

Component	Function
ALU	Performs mathematical and logical operations.
CU	Controls and coordinates all computer operations.

Memory Registers Temporarily store data and instructions.

Evolution (Generations) of the CPU

The CPU has evolved through **four generations**.

First Generation (1940–1956)

Technology

Vacuum Tubes

Features

- Very large (room-sized)
- Consumed a lot of electricity
- Produced a large amount of heat
- Required cooling systems
- Very expensive
- Speed measured in **milliseconds**

Examples

- ENIAC
- UNIVAC
- EDVAC

Second Generation (1956–1963)

Technology

Transistors

Features

- Smaller than first generation
- Consumed less electricity
- Produced less heat
- Faster than first generation
- Still expensive
- Speed measured in **microseconds**

Examples

- IBM 7030
- CDC 1604

Third Generation (1964–1971)

Technology

Integrated Circuits (IC)

Features

- Smaller computers
- Less power consumption
- Lower cost
- Faster performance
- Speed measured in **nanoseconds**

Examples

- IBM 360
 - CDC 6600
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Fourth Generation (1971 – Present)

Technology

Microprocessors

Features

- Very small size
- Very fast processing
- Low power consumption
- Less expensive
- Used in modern computers
- Speed measured in **MIPS** and **picoseconds**

Examples

- Modern desktop computers
- Laptops
- Tablets

Comparison of Computer Generations

Generation	Technology	Size	Speed	Power Consumption	Examples
First	Vacuum Tubes	Very Large	Milliseconds	Very High	ENIAC, UNIVAC
Second	Transistors	Smaller	Microseconds	Lower	IBM 7030
Third	Integrated Circuits	Smaller	Nanoseconds	Lower	IBM 360
Fourth	Microprocessors	Smallest	MIPS / Picoseconds	Lowest	Modern Computers

Time Measurement Units

Unit	Value
1 Millisecond (ms)	0.001 second
1 Microsecond (μs)	0.000001 second
1 Nanosecond (ns)	0.000000001 second
1 Picosecond (ps)	0.000000000001 second

CPU Speed (Clock Speed)

The **clock speed** of a CPU is the **number of instructions it can execute in one second**.

Unit of Measurement

Hertz (Hz)

Modern CPUs are measured in:

- **Kilohertz (kHz)**
 - **Megahertz (MHz)**
 - **Gigahertz (GHz)**
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Hertz Conversion

Unit	Value
1 kHz	1,000 Hz
1 MHz	1,000,000 Hz
1 GHz	1,000,000,000 Hz

Key Points to Remember

- CPU is called the **brain of the computer**.
- CPU processes data and executes instructions.
- CPU is located **on the motherboard inside the system unit**.
- CPU has **three main components**:
 - Arithmetic and Logic Unit (ALU)

- Control Unit (CU)
 - Memory Registers
- ALU performs mathematical and logical operations.
- CU controls all computer activities.
- Memory Registers temporarily store data and instructions.
- Computers are classified into **four generations** based on CPU technology:
 1. Vacuum Tubes
 2. Transistors
 3. Integrated Circuits
 4. Microprocessors
- CPU speed is measured using **Hz, MHz, and GHz**.
- As computer generations evolved, CPUs became **smaller, faster, cheaper, and more energy-efficient**.